

Discrete Mathematics (Final Exam)

April 30, 2026, 9.30 to 12:30

Total: 120 marks

Answer all questions.

1. A *boolean formula* is defined inductively: for every boolean variable x , both x and $\bar{x}$ are boolean formulas. If f and g are boolean formulas then so are $f \wedge g$, $f \vee g$, $\bar{f}$ and (f) . Let $\text{var}(f)$ denote the finite set of variables occurring in the boolean formula f . Given an assignment $\text{val} : \text{var}(f) \rightarrow \{0, 1\}$ of boolean values to the variables in f , the boolean evaluation of f is inductively defined. A formula f is *satisfiable* if there is a boolean assignment such that f evaluates to 1. Suppose X is an infinite set of boolean variables and Γ is a collection of boolean formulas f such that $\text{var}(f) \subset X$ for all $f \in \Gamma$. We say that Γ is *finitely satisfiable* if for every finite subset $\{f_1, f_2, \dots, f_r\}$ of Γ there is a boolean assignment to X that satisfies every f_i , $1 \leq i \leq r$. Furthermore, Γ is said to be *satisfiable* if there is a boolean assignment to X that satisfies every $f \in \Gamma$. The goal of this problem is to show that if Γ is finitely satisfiable then Γ is satisfiable.
 - (a) Let $\mathcal{P}$ denote the set of all boolean formulas f with $\text{var}(f) \subset X$. Thus, $\Gamma \subseteq \mathcal{P}$. Consider the following subset of $2^{\mathcal{P}}$: $\mathcal{T} = \{\Phi \subseteq \mathcal{P} \mid \Gamma \subseteq \Phi \text{ and } \Phi \text{ is finitely satisfiable}\}$. Show that $\mathcal{T}$ is a poset under set inclusion and that by Zorn's lemma $\mathcal{T}$ has a maximal element Ψ .
 - (b) Show that for every boolean formula $f \in \mathcal{P}$ either $f \in \Psi$ or $\bar{f} \in \Psi$. In particular, for every variable $x \in X$ either $x \in \Psi$ or $\bar{x} \in \Psi$.
 - (c) For each variable $x \in X$ define the boolean assignment $x \mapsto 1$ if $x \in \Psi$ and $x \mapsto 0$ otherwise. Show, using induction on formulas that this boolean assignment satisfies every formula in Ψ . Hence conclude that Γ is satisfiable.

25 marks

2. Use the inclusion-exclusion principle to prove that

$$n! = \sum_{k=0}^n (-1)^k \binom{n}{k} (n-k)^n.$$

10 marks

3. For a prime p , let $\mathbb{F}_p$ denote the finite field with p elements. Prove that the number of k -dimensional subspaces of the vector space $\mathbb{F}_p^n$ is given by

$$\begin{bmatrix} n \\ k \end{bmatrix}_p = \prod_{i=0}^{k-1} \frac{p^{n-i} - 1}{p^{k-i} - 1}.$$

10 marks

4. Determine the cycle index polynomial for coloring the edges of a regular tetrahedron that can be rotated in 3-dimensional space. How many distinguishable ways are there of r -coloring the edges of a regular tetrahedron? **10 marks**

5. Let G be a subgroup of S_n and consider the set $\mathcal{F}$ of all functions from $[n] \rightarrow [m]$.

- (a) Define the natural action of G on the set $\mathcal{F}$ and prove that it is a group action.
- (b) When does a permutation $\pi \in G$ fix a function $f \in \mathcal{F}$?
- (c) What is the number of orbits defined by G 's action on $\mathcal{F}$? Express it using the cycle index polynomial of G .
- (d) Let $n = 6$ and $m = 2$, and let G be the cyclic subgroup of S_6 generated by $(1\ 2\ 3)(5\ 6)$. What is the number of G -orbits in this case?

20 marks

6. Suppose $G = (V, E)$ is a planar graph without any triangles. Show that $|E| \leq 2|V| - 4$. How does this bound generalize to planar graphs that have no cycles of length less than g ? Justify with proof. **10 marks**

7. Let $K = \{k_1, k_2, \dots, k_r\} \subseteq [n]$. Consider the subset $\mathcal{P}$ of the power set $2^{[n]}$ defined as

$$\mathcal{P} = \{A \subseteq [n] \mid |A| \in K\}.$$

- (a) Let $\mathcal{A}$ be any antichain in the poset $(\mathcal{P}, \subseteq)$. Show that

$$\sum_{A \in \mathcal{A}} \frac{1}{\binom{n}{|A|}} \leq 1.$$

- (b) Hence show that every antichain in $(\mathcal{P}, \subseteq)$ has size bounded by $\max_{1 \leq i \leq r} \binom{n}{k_i}$.

15 marks

8. Suppose $G = (V, E)$ is an n -vertex graph that has no 4-cycle.

- (a) For $v \in V$ let $d(v)$ denote its degree. Show that the number of paths of length 2 in G with v as the middle vertex is $\binom{d(v)}{2}$.
- (b) G has a 4-cycle if and only if there are four distinct vertices u, v, x, y such that x, u, y and x, v, y are paths of length 2. Hence show that if G has no 4-cycles then

$$\binom{d(v)}{2} \leq \binom{n}{2}.$$

- (c) Using the above inequality prove that $|E| \leq cn^{3/2}$ for some constant $c > 0$.

20 marks